King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Major Exam II Term 232

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. If $f(x) = \dfrac{x+6}{x-2}$ and $x > 2$, then $(f^{-1})'(3) =$

(a) 2

(b) -4

(c) -2

(d) 6

(e) 4

2. The derivative of $y = \dfrac{e^x}{2}(\cos x + \sin x)$ is

(a) $e^x \cos x$

(b) $\dfrac{e^x}{2}(\cos x + \sin x)$

(c) $e^x \sin x$

(d) $\dfrac{e^x}{2}(-\sin x + \cos x)$

(e) 0

3. An airplane is flying at an altitude of 6 kilometers on a flight path that will take it directly over a radar tracking station. If the distance S between the airplane and the radar station is decreasing at a rate of 400 kilometers per hour when $S = 10$ kilometers, then the speed of the plane is

(a) 450 km/h

(b) 550 km/h

(c) 500 km/h

(d) -550 km/h

(e) -600 km/h

4. Using Newton's Method if the initial guess to approximate the zero of $f(x) = 2x^2 - 1$ is $x_1 = \frac{1}{2}$, then $x_3 =$

(a) $\frac{13}{21}$

(b) $\frac{17}{24}$

(c) $\frac{11}{16}$

(d) $\frac{19}{24}$

(e) $\frac{5}{8}$

5. The value of c in the interval $(1, 9)$ that satisfies the conclusion of the Mean Value Theorem for $f(x) = \sqrt{x}$ is

(a) 5

(b) 6

(c) 4

(d) 3

(e) 2

6. If $f(x) = \frac{2 - \frac{1}{x}}{x - 3}$, then $f'(2) =$

(a) -2

(b) $\frac{5}{4}$

(c) $\frac{3}{2}$

(d) $-\frac{3}{2}$

(e) $-\frac{7}{4}$

7. The voltage V in volts of an electrical circuit is $V = IR$, where R is the resistance in ohms and I is the current in amperes. R is increasing at a rate of 2 ohms per second, and V is increasing at a rate of 8 volts per second. At what rate is I changing when $V = 10$ volts and $R = 5$ ohms?

(a) 1/5 amperes per second

(b) 3/5 amperes per second

(c) 2/5 amperes per second

(d) 8/5 amperes per second

(e) 4/5 amperes per second

8. The slope of the tangent line to the curve $y = \dfrac{1+\sec x}{1-\sec x}$ at the point $\left(\dfrac{\pi}{3}, -3\right)$ is equal to

(a) 8

(b) $-\sqrt{3}$

(c) $4\sqrt{3}$

(d) -8

(e) $-4\sqrt{3}$

9. The equation of the tangent line to the curve $\arctan(xy) = \arcsin(x+y)$ at $(0,0)$ is

(a) $y = x$

(b) $y = \pi x$

(c) $y = -x$

(d) $y = \dfrac{\pi}{4}x$

(e) $y = 2x$

10. The slope of the normal line to the curve $\sin(\pi xy) = \ln x$ at the point $(1,0)$ is equal to

(a) $-\pi$

(b) -2π

(c) $-\dfrac{e}{\pi}$

(d) $-\dfrac{1}{\pi e}$

(e) $-\dfrac{1}{2\pi}$

11. The sum of the extreme values of the function $f(x) = 1 + 4x - 6x^{2/3}$ on the interval $[-1,1]$ is equal to

(a) -10

(b) 0

(c) -7

(d) -8

(e) -9

12. If $g(x) = \ln\left(\dfrac{3x^2-2}{2x+3}\right)^{-2}$, then $g'(-1) =$

(a) 8

(b) -4

(c) -8

(d) 16

(e) -16

13. If $x + y - 1 = \ln(x^2 + y^2)$, then $\dfrac{d^2y}{dx^2}$ at the point $(1, 0)$ is equal to

(a) -1

(b) 1

(c) 2

(d) 0

(e) -2

14. If $y = \arcsin x + x\sqrt{1-x^2}$, then $\dfrac{dy}{dx} =$

(a) $2\sqrt{1-x^2}$

(b) $-2\sqrt{1-x^2}$

(c) $2\sqrt{x^2-1} - 1$

(d) $\dfrac{-1}{\sqrt{1-x^2}} - \dfrac{x^2}{\sqrt{1-x^2}} + \sqrt{1-x^2}$

(e) $\dfrac{1}{\sqrt{x^2-1}} + \dfrac{x^2}{\sqrt{1-x^2}} - \sqrt{1-x^2}$

15. If the graph of $f(x) = x^2 - x - 2$ intersects the x-axis at a and b where $a < b$ and if c is the value between a and b that satisfies Rolle's Theorem, then $a + b + c =$

(a) $\dfrac{1}{2}$

(b) 1

(c) $\dfrac{5}{2}$

(d) $\dfrac{3}{2}$

(e) 2

16. If $f(x) = 2\sin x - \cos 2x$, then the absolute maximum of f over the interval $[0, 2\pi]$ is equal to

(a) $-\frac{3}{2}$

(b) $\frac{3}{2}$

(c) 6

(d) -1

(e) 3

17. If (a, b) is a point on the graph of $f(x) = \frac{x+1}{x-1}$ at which the tangent line is parallel to the line $2y + x = 6$, then $a + b =$

(a) 0 or 3

(b) -2 or -3

(c) 2 or 3

(d) -1 or 5

(e) 1 or -5

18. If $x > 0$, $y > 0$ and $y = x^{xy}$, then $\frac{dy}{dx} =$

(a) $\frac{y^2(1 + 2\ln x)}{1 - 2xy\ln x}$

(b) $\frac{y^2(1 + x\ln x)}{1 - y\ln x}$

(c) $\frac{y^2(1 - \ln x)}{1 + xy\ln x}$

(d) $\frac{y(1 + \ln x)}{1 - xy^2\ln x}$

(e) $\frac{y^2(1 + \ln x)}{1 - xy\ln x}$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$f(6)=3$, $f'(6)=-\frac{1}{2}$, so $(f^{-1})'(3)=-2$.
2	**(a)**	Product rule: $\frac{e^x}{2}(\cos x+\sin x)+\frac{e^x}{2}(\cos x-\sin x)=e^x\cos x$.
3	**(c)**	$S^2=x^2+36$, $x=8$: $10(-400)=8x'$, $x'=-500$, speed 500 km/h.
4	**(b)**	$x_{n+1}=x_n-\frac{2x_n^2-1}{4x_n}$: $x_2=\frac{3}{4}$, $x_3=\frac{17}{24}$.
5	**(c)**	$f'(c)=\frac{1}{2\sqrt{c}}=\frac{3-1}{8}=\frac{1}{4}$, so $c=4$.
6	**(e)**	Quotient rule on $\frac{2x-1}{x^2-3x}$: $f'(2)=\frac{-4-3}{4}=-\frac{7}{4}$.
7	**(e)**	$I=2$; $V'=I'R+IR'$: $8=5I'+4$, so $I'=\frac{4}{5}$ A/s.
8	**(c)**	$y'=\frac{2\sec x\tan x}{(1-\sec x)^2}$; at $\frac{\pi}{3}$: $\frac{4\sqrt{3}}{1}=4\sqrt{3}$.
9	**(c)**	At $(0,0)$: $0=1+y'$, so slope -1: $y=-x$.
10	**(a)**	$\pi\cos(\pi xy)(y+xy')=\frac{1}{x}$ gives $y'=\frac{1}{\pi}$; normal slope $-\pi$.
11	**(d)**	$f(-1)=-9$, $f(0)=1$, $f(1)=-1$: max 1, min -9, sum -8.
12	**(d)**	$g=-2[\ln(3x^2-2)-\ln(2x+3)]$, $g'(-1)=-2(-6-2)=16$.
13	**(d)**	$y'(1,0)=1$; second differentiation gives $y''=\frac{4\cdot1-4}{1}=0$.
14	**(a)**	$y'=\frac{1}{\sqrt{1-x^2}}+\sqrt{1-x^2}-\frac{x^2}{\sqrt{1-x^2}}=2\sqrt{1-x^2}$.
15	**(d)**	$a=-1$, $b=2$, $f'(c)=2c-1=0$ gives $c=\frac{1}{2}$; sum $\frac{3}{2}$.
16	**(e)**	Critical values: 3 at $\frac{\pi}{2}$, -1, $-\frac{3}{2}$; absolute max $=3$.
17	**(d)**	Slope $-\frac{1}{2}$: $\frac{-2}{(x-1)^2}=-\frac{1}{2}$, $x=3,-1$; points $(3,2),(-1,0)$: 5 or -1.
18	**(e)**	$\ln y=xy\ln x$, differentiate and solve: $y'=\frac{y^2(1+\ln x)}{1-xy\ln x}$.